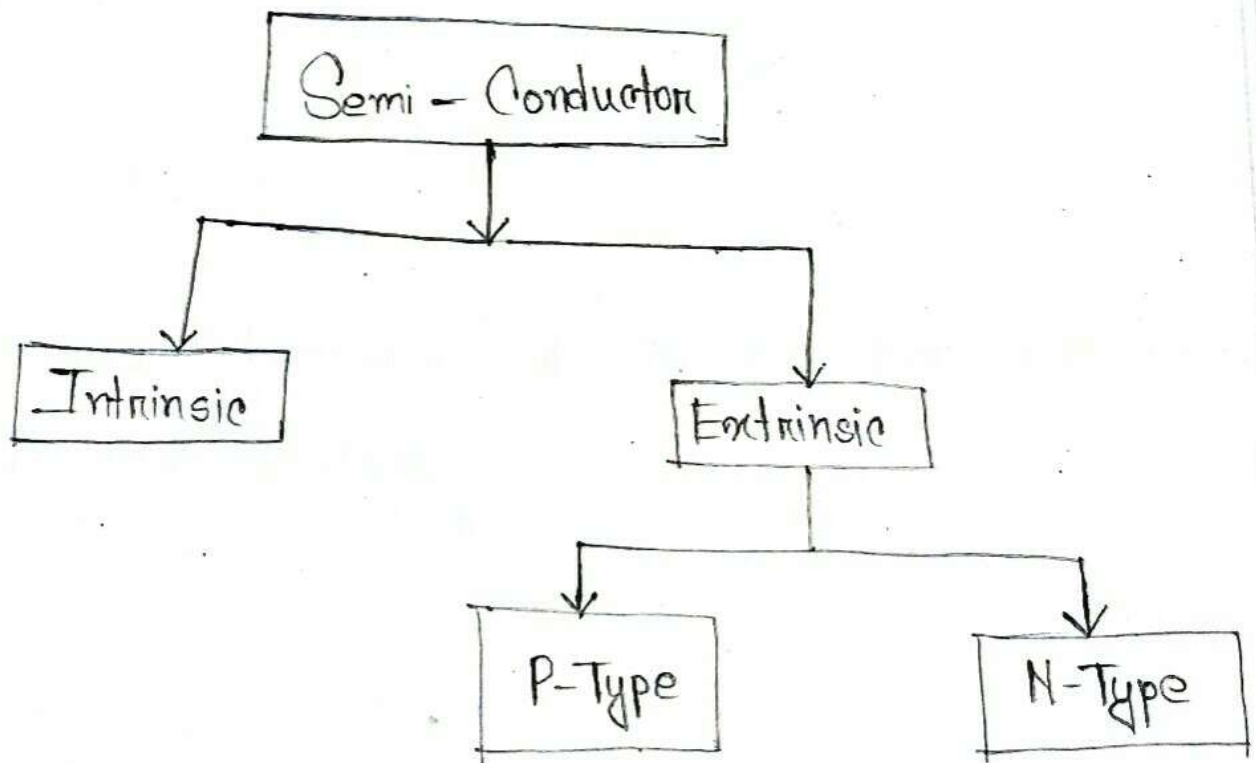


What is semiconductor? Classify the semiconductor material?

Semiconductors are materials whose electrical properties lie between Conductors and Insulators. Common elements such as carbon, silicon and germanium are semiconductors. Silicon is the best and most widely used semiconductor.



☐ Define intrinsic semiconductor, or, undoped semiconductor or i-type semiconductor?

Intrinsic Semiconductor : An intrinsic semiconductor is a semiconductor that is chemically pure and free of impurities. It is also known as undoped semiconductor, pure semiconductor or i-type semiconductor.

☐ Define Extrinsic semiconductor, or, doped semiconductor?

Extrinsic Semiconductor : An extrinsic semiconductor is a semiconductor that has been intentionally doped with impurities to change its electrical properties. The process of adding impurities is called as Doping.

Q. Difference between intrinsic semiconductor and extrinsic semiconductor?

Intrinsic Semiconductor

i) An intrinsic semiconductor is a semiconductor that is chemically pure and free of impurities.

ii) Its electrical conductivity is low.

iii) The number of excited electron and the number of holes are equal : $n = p$

Extrinsic Semiconductor

i) An extrinsic semiconductor is a semiconductor that has been intentionally doped with impurities to change its electrical properties.

ii) Its electrical conductivity is high.

iii) The number of free electrons and holes is never equal.

iv) The electrons and holes are solely created by thermal excitation.

iv) There is excess of electrons in n-type semiconductors and excess of holes in p-type semiconductors.

v) It is known as undoped or i-type or pure semiconductor.

v) It is known as doped or impure semiconductor.

vi) No impurity atoms are added to it.

vi) Impurity atoms doped the structure.

vii) These are not practically used.

vii) These are practically used.

viii) There is no classification of intrinsic semiconductor.

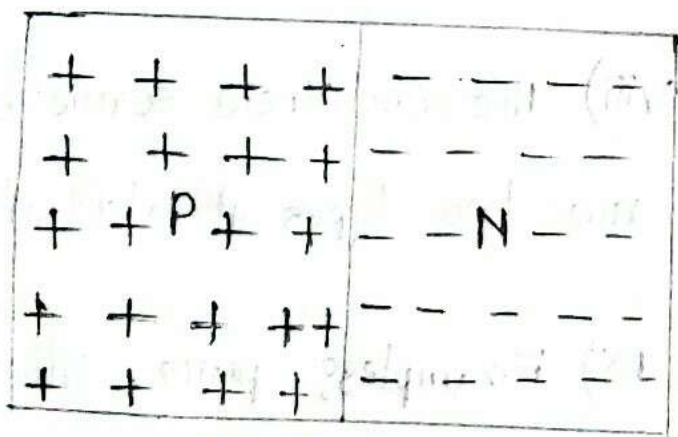
viii) The extrinsic semiconductors are two types of classification.

ix) Examples: pure silicon and germanium.

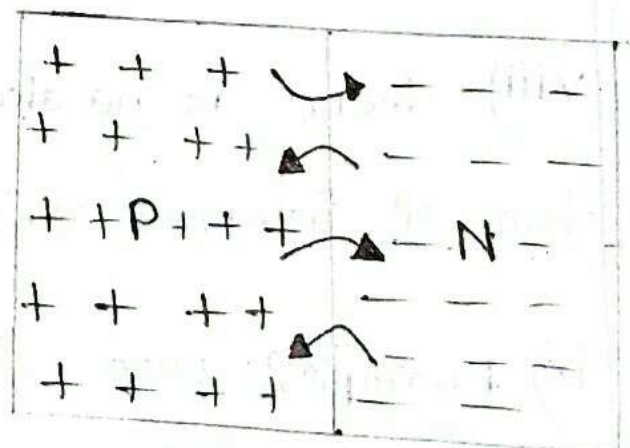
ix) Examples: pure silicon and germanium doped with impurity atoms of As, P, Bi, Sb etc.

Q. What is the formation of PN Junction?

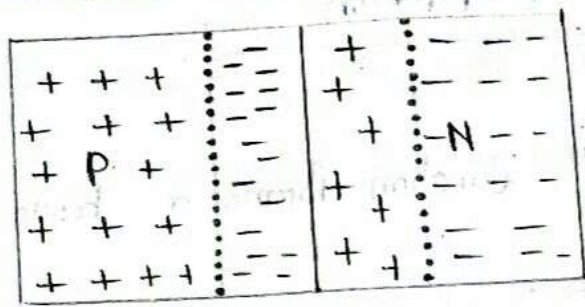
P-n junction are formed by joining n-type and p-type semiconductor materials, as shown below. The P side is called as anode and N side is called cathode. Since the n-type region has a high electron concentration and the p-type a high hole concentration, electrons diffuse from the n-type side to the p-type side. Similarly, holes flow by diffusion from the p-type side to the n-type side.



PN Junction



Diffusion



Depletion Layer

When a p-type semiconductor material is suitably joined to n-type semiconductor the contact surface is called p-n junction.

Definition

Diode : The PN Junction forms a basic semiconductor device called diode.

Biasing : If an external dc supply voltage is applied to any electronic device, it is called biasing.

Depletion layer : An electric field $\hat{E}$ forms between the positive ion cores in the n-type material and negative ion cores in the p-type material. This layer is called depletion layer.

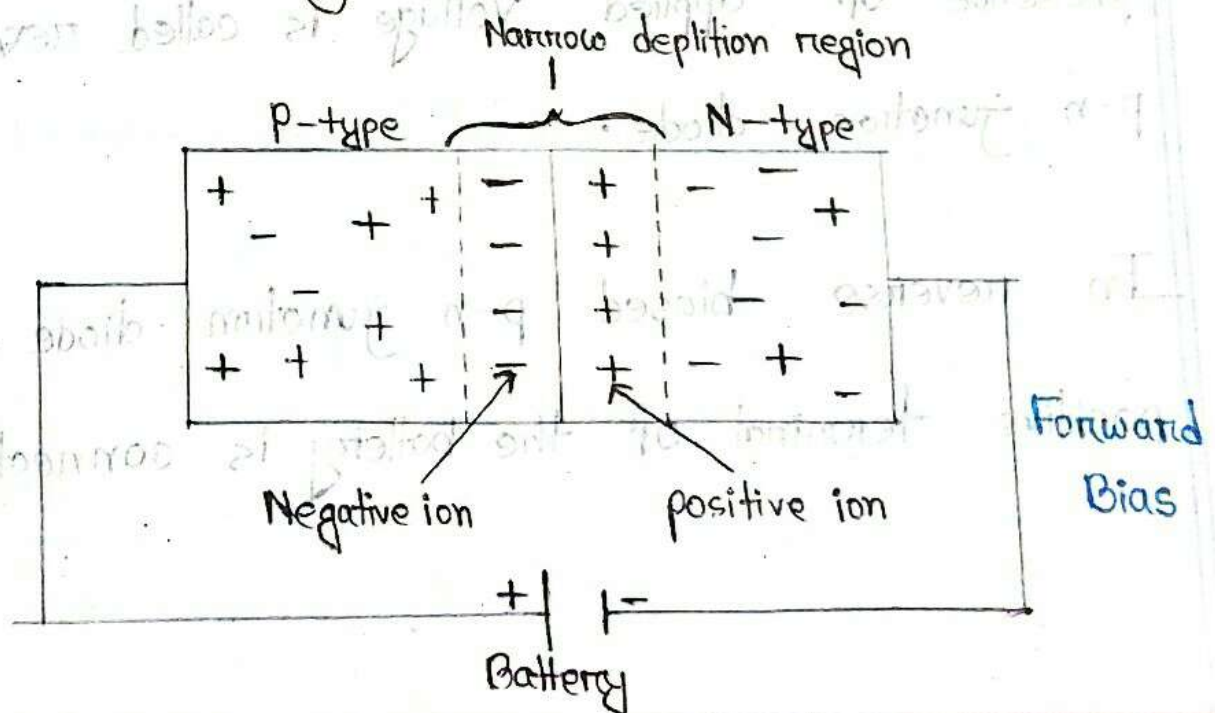
Diffusion : Diffusion is the net movement of anything (for example atoms, ions, energy) generally from a region of higher concentration to a region of lower concentration.

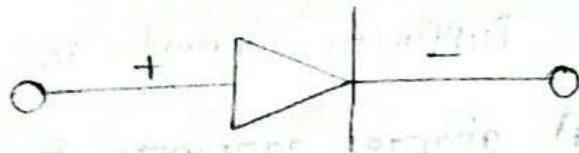
Diffusion Current : Diffusion current is a current in or is the flow of charge carriers in a semiconductor from a region of higher concentration to a region of lower concentration.

Drift Velocity : Drift velocity is the average velocity of charged particles in a material when an electric field is applied. SI unit is $(m/s)/(m/s)$.

Forward Bias : When, connect p-type region

of a junction with the positive terminal of a voltage source and n-type region with the negative terminal of the voltage source, then the junction is called forward bias;





Forward Biased



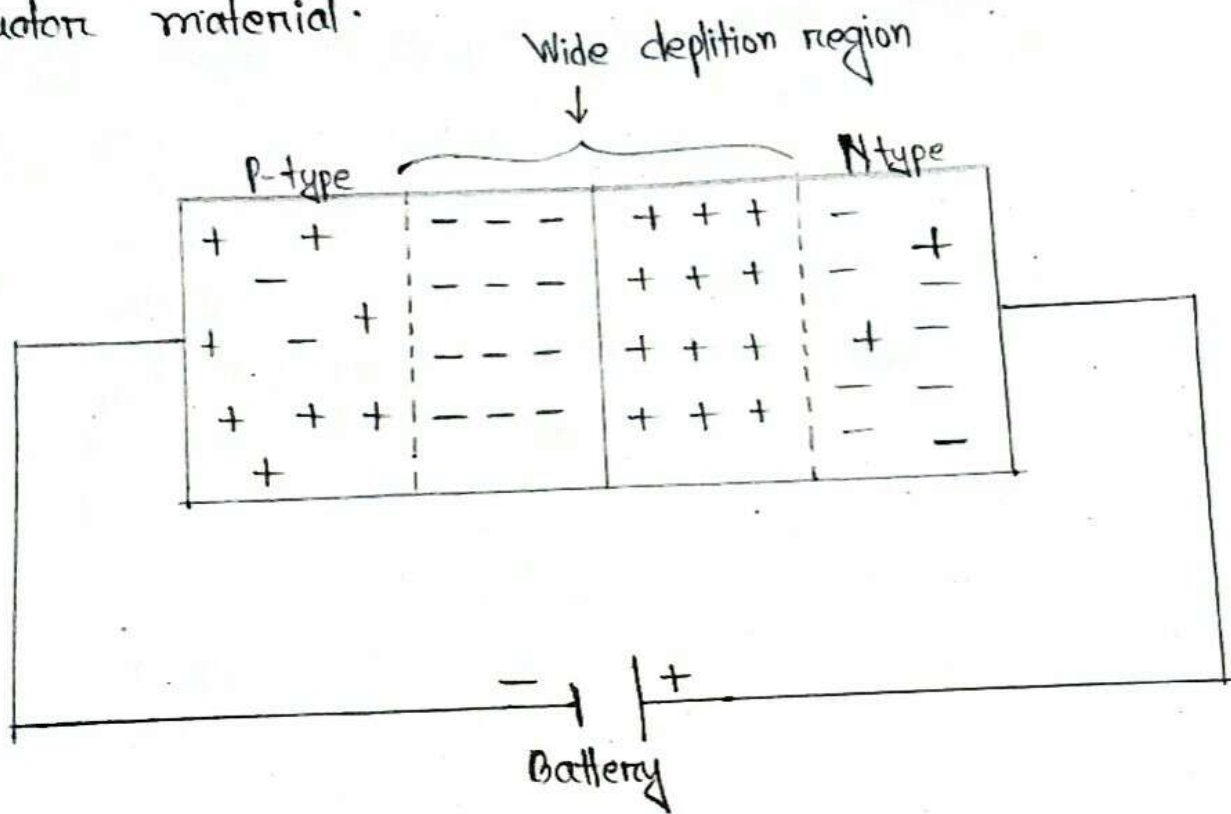
Forward Bias (switch closed)

Reverse Bias :

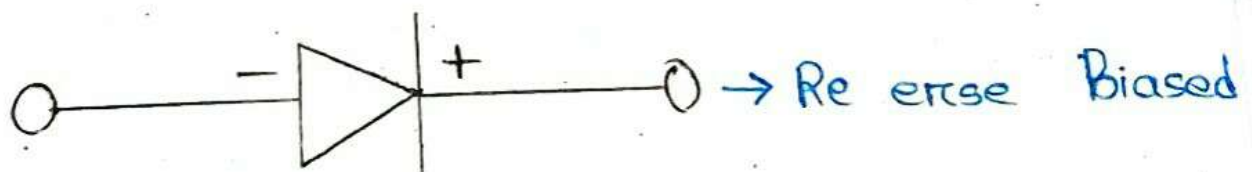
The process, when a p-n junction diode blocks the electric current in the presence of applied voltage is called reverse biased p-n junction diode.

In reverse biased p-n junction diode, the positive terminal of the battery is connected to the

n-type semiconductor material and the negative terminal of the battery is connected to the p-type semiconductor material.



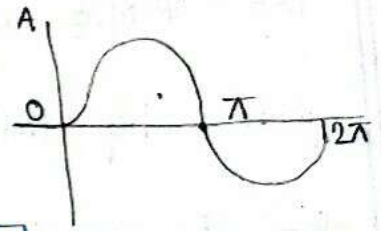
Reverse bias



Reverse Bias (switch open)

Average value :

$$i = I_m \sin \omega t = I_m \sin \theta$$



$$I_{avg} = \int_0^{\pi} \frac{i d\theta}{\pi - 0} \quad [\because (2\pi - \pi) \text{ হতে পারে}]$$

Putting $i = I_m \sin \theta$

[$\because$ average value -
তে negative wave
ধরা হয় না]

$$I_{avg} = \int_0^{\pi} \frac{I_m \sin \theta}{\pi} d\theta$$

$$= \frac{I_m}{\pi} \int_0^{\pi} \sin \theta d\theta$$

$$= \frac{I_m}{\pi} [-\cos \theta]_0^{\pi}$$

$$= \frac{I_m}{\pi} [-\cos \pi + \cos 0]$$

$$\therefore = \frac{I_m}{\pi} (1+1) = \frac{2I_m}{\pi}$$

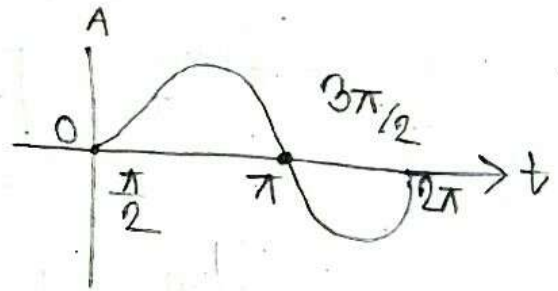
$$\therefore I_{avg} = 0.637 I_m$$

R.M.S value :

The R.M.S value of an alternating current is given by that (d.c) current which when flowing through a given circuit for a given time produces the same heat as produced by the alternating current when flowing through the same circuit for the same time.

$$I = I_m \sin \omega t = I_m \sin \theta$$

$$= \int_0^{2\pi} \frac{I^2 d\theta}{2\pi - 0}$$



The square root of this value is, $\left[\because \text{full area} \right]$

$$I = \sqrt{\int_0^{2\pi} \frac{I_m^2}{2\pi - 0} d\theta}$$

$$= \sqrt{\int_0^{2\pi} \frac{I_m^2 \sin^2 \theta}{2\pi - 0} d\theta}$$

$$= \sqrt{\frac{I_m^2}{4\pi} \int_0^{2\pi} \frac{2 \sin^2 \theta}{0.1} d\theta}$$

$$= \sqrt{\frac{I_m^2}{4\pi} \int_0^{2\pi} (1 - \cos 2\theta) d\theta}$$

$$= \sqrt{\frac{I_m^2}{4\pi} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{2\pi}}$$

$$= 0.707 I_m$$

$$= \frac{I_m}{4\sqrt{\pi}} \sqrt{\left[2\pi - \frac{\sin 4\pi}{2} \right]}$$

$$= \frac{I_m}{4\sqrt{\pi}} \sqrt{2\pi}$$

$$\text{Form factor} = \frac{\text{r.m.s value}}{\text{average value}}$$

$$= \frac{0.707 I_m}{0.637 I_m}$$

$$\therefore \text{FF, } k_a = 1.11$$

Form factor is the ratio of the root mean square (RMS) value of a waveform to its average.

Amplitude factor / Peak factor / Crest factor

⇒ Amplitude factor is defined as the ratio of maximum value of the R.M.S value of an alternating quantity.

$$\text{CF, } k_p = \frac{\text{Maximum value}}{\text{r.m.s value}} = \frac{I_m}{0.707 I_m} = 1.414$$